Reflection

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I picked Project 3 for my Final for a number of reasons. For one, I was unable to provide a completed assignment simply because I have had many issues with trying to make it available via a webpage link. For that reason, I knew that this was a good chance to show that I understood the assignment and was able to provide something for it. The other reason why I wanted to use this project was because of what it can be used for by students. Students being able to visualize their graphs differentiability as well as have complete control of the graph to experiment multiple what-if scenarios was quite intriguing. Current applications like Desmos and Geogebra work similarly, however I never have used them for this very purpose. As someone going into the education field, I wanted to really test my skill in creating a site or application that my future students may find helpful for their studies.

This project showcases skills in differentiability of a function. By creating an application visualizing this very skill, I was able to see for myself how there are limitations that numerical calculation of a function can have some gaps when determining limits and concavity. The visualizer allows the user to use different sliders to adjust the graph in real time to help compare their calculations to the actual graph on display.

I noticed that the limitations of the app include the determining that the function is not differentiable when simply the derivative would just be inconclusive. A function’s derivative being inconclusive does not necessarily mean that the function itself is not differentiable. To adjust my original code, I instructed AI to help with this and provide a new, updated code to fix this problem.

Some skills I have gained from this is being able to more easily see where applications can have limitations. The aforementioned app limitation is a perfect example of this. Now knowing how to find issues within an app, I can use AI to help fix the code by giving it more exact and precise instructions on what I need adjusted. This allows me to be able to have a cleaner design and more accurate productions.

This portfolio I feel will help my professional goals in multiple ways. The first is that my ability to communicate with AI with more precision is greatly enhanced. In a world where the education of students is being heavily orchestrated through AI content and assistance, I feel that with the skills I have gained, Ill be able to quickly recognize any shortcomings that AI will have and adjust them to more accurately get what is needed for the students' lesson plans and assignments. Another reason this portfolio helps is by

showcasing what I have learned. By doing so, I can gain feedback from others in the field to help me adjust and improve the site as well as collaborate with other instructors on developing new software for the future generations.